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Benchmarking Highly Precise Quantum Linear Systems Algorithms

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Abstract

Systems of linear equations are ubiquitous across science, engineering, machine learning, and even finance. While classical methods can be prohibitively slow for large-scale problems, quantum linear systems algorithms offer the potential for exponential speedup in certain parameter regimes. However, a significant gap persists between this theoretical promise and practical implementation, as the advantages are often obscured by the substantial quantum resources and high sensitivity to noise inherent in current quantum hardware. One way to bridge this gap is through the use of Iterative Refinement, a classical post-processing scheme that can exponentially improve the accuracy to which a linear system of equations can be solved using low-precision arithmetic. In the context of quantum linear systems algorithms, such as the HHL algorithm proposed by Harrow, Hassidim, and Loyd, Iterative Refinement can greatly reduce the quantum resources required to calculate an accurate solution in terms of tomography cost, circuit volume, and fault-tolerant overhead. Here, we compute and benchmark highly precise solutions to linear systems of equations of up to eight variables by running HHL with Iterative Refinement on NISQ quantum computers. We also present our open-source implementation, emphasizing that our circuit is not tailored to specific problem instances, as most available implementations are.

Index Terms

Quantum Linear System Algorithm, HHL Algorithm, Iterative Refinement, Benchmarking

Poster relevance This paper focuses on the implementation of quantum linear system algorithms, with improvements achieved through iterative refinement. The work is fully aligned with the themes of the IEEE Quantum Week conference and falls under the categories of Quantum Algorithms & Information and Quantum Computing. It is directly relevant to several focus areas and has potential applications across others, including hybrid quantum-classical computing architectures & algorithms, quantum machine learning (QML), quantum benchmarking, quantum optimization techniques, and quantum applications.

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I. Introduction

It is widely believed that quantum computers can efficiently solve some problems that do not admit efficient classical algorithms. Recently, a considerable amount of attention has been devoted to quantum linear algebra. A common class of methods use Hamiltonian simulation to prepare a quantum state $|x\rangle$ that is proportional to the solution of the linear system

$$Ax = b, \tag{1}$$

for a given Hermitian matrix $A \in \mathbb{R}^{n\times n}$ and a vector $b \in \mathbb{R}^n$. In the *quantum linear systems problem* (QLSP), one has input oracle access to the entries of A and the ability to prepare a quantum state $|b\rangle$ that is proportional to the right-hand side vector b. Research into this subfield began with the work of Harrow, Hassidim, and Lloyd [1], who proposed what has come to be known as the HHL algorithm for solving the QLSP.

Many linear systems problems (for example, solving the Newton system within an interior point method algorithm for linear or semidefinite optimization [2]) require high-precision solutions, which can be challenging to compute on imperfect hardware. Iterative Refinement (IR) is a classical technique originally developed in response to this challenge on low-precision classical computers. IR can use an inexact or low-precision solution as a starting point and systematically improve its accuracy [3], [4]. This idea has since been extended to compute highly accurate (and at times, *exact*) solutions to linear [5] and semidefinite optimization problems [6], and recent works have investigated their adaptability to quantum algorithms [7], [8]. Given the speedups IR has facilitated for optimization, it is worthwhile to investigate how IR can be leveraged in combination with QLSAs. Here we show that both in theory and in practice, by using a NISQ device to solve for the residual error term at each step, we can achieve high-precision final solutions while placing significantly fewer demands on the quantum hardware.

The iterative refinement method for solving linear system of equations [3] is described in Algorithm 1, and we summarize the core idea as follows. The algorithm commences from an initial solution $x^{(0)} \in \mathbb{R}^d$, and subsequently computes a refined solution $x^{(k+1)} \leftarrow x^{(k)} + u^{(k)}$ in iterations $k = 0, 1, 2, \ldots$, where $u^{(k)}$ acts as a correction of the error $r^{(k)} = z - Mx^{(k)}$ and is the solution to the *refining system* $Mu^{(k)} = r^{(k)}$. These operations can all be carried out using the same level of accuracy or *fixed precision*.

A. Contributions

Our primary contribution is a benchmarking suite, available at https://github.com/QCOL-LU/QLSAs, for running the HHL algorithm with iterative refinement on quantum computer simulators/emulators as well as on real hardware. The suite is integrated with the Quantinuum qnexus library for seamless execution on supported backends, including Quantinuum's own H-series hardware and high-performance emulators. Our implementation is generalized (not hard-coded to specific instances or sizes), unlike the majority of available implementations found online.

We also demonstrate that classical IR post-processing significantly lowers the quantum resource requirements for HHL. By iteratively refining a low-precision quantum solution, we reduce the need for high-precision quantum phase estimation (QPE). This directly translates to:

- Shallower circuits with fewer gates.
- Fewer qubits required for the QPE register.
- A reduced number of shots needed per execution.
- Viable execution on current NISQ hardware, effectively enabling a fault-tolerant-era algorithm to provide meaningful results today.

Lastly, we benchmark results for systems up to 4 variables on Quantinuum H2-2, and 8 variables on various emulators/simulators, quantifying the resource reductions and demonstrating exponential improvements in solution accuracy even on real noisy hardware.

II. Algorithm Implementation

Here, we discuss the details of our HHL-IR implementation.

A. Iterative Refinement

Our implementation follows the fixed-precision IR scheme detailed in and summarized in Algorithm 1, which alternates between a quantum solver and classical post-processing. This approach uses the quantum computer only for the most challenging step (solving the linear system), while leveraging classical resources for all scaling and update operations.

Algorithm 1 Iterative Refinement for the Linear Systems Problem

Input: Coefficient matrix $A \in \mathbb{R}^{d\times d}$ with $\left\|A^{-1}\right\| \le \kappa_A$, right-hand side vector $b \in \mathbb{R}^d$, error tolerances $0 < \zeta \ll \xi < 1$.
Output: A classical solution vector $x \in \mathbb{R}^d$ satisfying $\left\|Ax - \frac{b}{\|b\|}\right\| \le \zeta$.
Let $\tilde{\alpha} = \max\{\|A\|_F, \|b\|\}$. Then, normalize the system $\tilde{b} \leftarrow \frac{b}{\tilde{\alpha}}$ and $\tilde{A} \leftarrow \frac{A}{\tilde{\alpha}}$.
Choose starting point $(x^{(0)}, r^{(0)}, \eta^{(0)}) \leftarrow (0, \tilde{b}, 1)$, $k \leftarrow 0$.
while $\|r^{(k)}\| > \frac{\zeta}{\tilde{\alpha}}$
1) $\tilde{u}^{(k)} \leftarrow$ **solve** $(A, \eta^{(k)} r^{(k)})$ using $O_{LS}(\xi)$
2) $x^{(k+1)} \leftarrow x^{(k)} + \frac{1}{\eta^{(k)}}\tilde{u}^{(k)}$
3) $r^{(k+1)} \leftarrow \tilde{b} - Ax^{(k+1)}$
4) $\eta^{(k+1)} \leftarrow \frac{1}{\|r^{(k+1)}\|}$
5) $k \leftarrow k + 1$
end

B. HHL Circuit and Execution

The core quantum component is the HHL algorithm, which we implement in Qiskit. The circuit consists of three main stages: Quantum Phase Estimation (QPE) to find the eigenvalues of the matrix A, a controlled rotation based on the inverted eigenvalues, and an inverse QPE to uncompute the register. For hardware execution on Quantinuum systems, circuits are transpiled using pytket and submitted via the qnexus library, which handles the full compilation and job management workflow.

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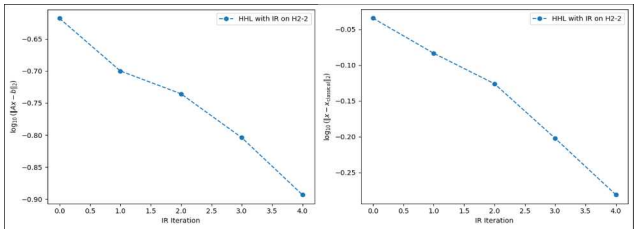

Fig. 1. Residuals and errors from a 4 variable linear system as a function of iterative refinement iteration on Quantinuum H2-2.

III. Experimental Results

We run HHL-IR on various backends, including both emulators/simulators and quantum hardware. The results demonstrate that iterative refinement consistently and significantly improves the quality of the solutions obtained from the HHL algorithm.

On quantum hardware, we are able to run system sizes up to 4 by 4. Figure 1 plots the log scale results for an unstructured 4-variable system executed on the Quantinuum H2-2 quantum computer. Despite the presence of hardware noise, IR systematically reduces the solution error and residual, improving the accuracy of the final result with each iteration.

A detailed summary of our benchmarking is presented in the table in Figure 2. This data quantifies the performance of HHL with and without IR across various problem sizes and backends. For every case, the application of IR yields a dramatic reduction in both the final solution error $||x_c - x_q||$ and the residual $||Ax - b||$. For example, for an 8-variable problem on the H1-1LE emulator, IR reduced the solution error by over three orders of magnitude, from approximately 0.8488 to 0.0008. These results empirically validate that IR is a crucial tool for obtaining high-precision solutions from quantum linear solvers on current and near-term hardware.

We also show that IR is the most resource efficient way to compute a high precision solution on current hardware. Figure 3 shows the residuals as a function of both IR iteration and the number of qubits used for the Quantum Phase Estimation (QPE) subroutine. Every additional qubit used in the QPE clock register, in theory, improves exponentially the precision to which the eigenvalues of the matrix A are estimated. However, circuit depth also increases exponentially. While a 4 by 4 linear system on a noisy emulator reproduces this theory, an 8 by 8 linear system actually gets worse with increasing QPE qubits due to the accumulation of noise in the circuit. On both system sizes, IR improves precision as expected.

IV. Conclusions

Quantum linear systems algorithms such as HHL can provide speedups over their classical counterparts. However, many instances of linear systems problems require solutions calculated to high precision. Although the requirements for achieving these levels of precision directly with QLSAs like the HHL algorithm are out of reach on today's hardware, integrating QLSAs with iterative refinement can exponentially improve the quality of the solution with linear cost in time and no cost in space. Here, we take a step toward bridging the gap between theory and practice, providing an open-source implementation with state of the art results in solution quality on current NISQ hardware.

Emulator Results

Backend	Problem Size	Condition Number	Sparsity	Number of Qubits	Circuit Depth	Total Gates	‖x_c - x_q‖ without IR	‖x_c - x_q‖ with IR	‖Ax - b‖ without IR	‖Ax - b‖ with IR	Total Iterations of IR
aer_simulator	2 x 2	1.872340	0.500000	4	21	25	0.379778	0.000000	0.327757	0.000000	5
H1-1LE	2 x 2	1.872340	0.500000	4	12	22	0.255828	0.000050	0.121943	0.000042	10
H2-1LE	2 x 2	1.872340	0.500000	4	12	22	0.281076	0.000092	0.132106	0.000051	10
aer_simulator	4 x 4	2.571542	0.625000	7	107	132	0.547150	0.000030	0.141385	0.000009	9
H1-1LE	4 x 4	2.571542	0.625000	7	323	449	0.711828	0.096980	0.189305	0.024518	10
H2-1LE	4 x 4	2.571542	0.625000	7	323	449	0.765401	0.056367	0.204540	0.014552	10
aer_simulator	8 x 8	2.232571	0.750000	12	1007	1093	0.771855	0.000142	0.258965	0.000055	10
H1-1LE	8 x 8	2.232571	0.750000	12	2997	4310	0.848844	0.000807	0.305588	0.000260	10
H2-1LE	8 x 8	2.232571	0.750000	12	2997	4310	0.857951	0.003080	0.309835	0.000992	10

Fig. 2. Example statistics gathered in our QLSA benchmarking suite on a variety of problem sizes and backends.

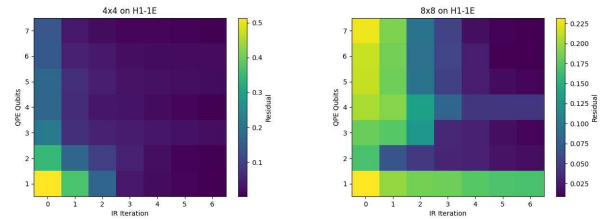

Fig. 3. Residual comparison for 4-variable (left) and 8-variable (right) systems across iterative refinement (IR) iterations and varying numbers of QPE qubits, obtained using the H1-1E noisy emulator.

Future work will focus on expanding this benchmarking suite to further enhance its practicality and scope. We plan to incorporate and evaluate other promising quantum linear systems algorithms, such as variational QLSAs, to provide a more comprehensive comparison of available methods. We will also extend backend support to a wider array of quantum platforms beyond Quantinuum and IBM, ensuring our findings are broadly applicable and robust across different hardware architectures.

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